

CS 30700: Project Charter

BoilerSpace

Team 3: Aagam Bapna, Sheehan Dandapat, Rohan Potta, Nathan Wan, Jed Macapinlac, Bidit Acharyya

1. Problem Statement

Purdue students struggle to find open study spaces in busy buildings like Lawson, WALC, or Krannert because there is no way to check room availability without physically walking to the location, sinking 30 to 60 minutes walking between buildings during peak hours. Additionally, students often miss out on helpful academic collaboration because there is no central hub to schedule future study sessions or access notes from previous group meetings. Our project, BoilerSpace, is different because it combines a live crowdsourced room map with a permanent digital library and scheduling tool. Unlike the Purdue room scheduling system that is generic and does not tailor a studying experience, or platforms like GroupMe that focus on real-time messaging that specific course collaboration, our platform allows users to search through past session notes using an AI chat, congregated by a multitude of students from the class, and a view “live vibes” of campus buildings to decide where to work with likeminded perspective students.

2. Project Objectives

- Develop an interactive digital map of Purdue campus buildings that displays real time room occupancy and availability based on student check ins along with what people are studying in each room.
- Build a scheduling system that allows students to create and advertise future study sessions for specific courses to improve peer collaboration.
- Allow Purdue student organizations to hold open study meetings/tutoring sessions to specific users of the application.
- Create a centralized knowledge repository where users can upload PDF notes or summaries from their study sessions for others to view later.
- Integrate a Generative AI assistant that can answer questions about the content of uploaded notes and recommend study spots based on a student's needs.
- Establish a professional software engineering workflow using the Scrum methodology, including regular sprint planning, CI/CD pipelining, quality testing, and retrospective documents

3. Stakeholders

Users: Purdue University students looking to discover study spaces, connect with peers, organize group meetings, and share academic resources, as well as student organizations and clubs that use the platform to schedule callouts and meetings.

Developers: Aagam Bapna, Sheehan Dandapat, Rohan Potta, Nathan Wan, Jed Macapinlac, Bidit Acharyya

Project Manager: Devansh Khandelwal

Project Owners: Aagam Bapna, Sheehan Dandapat, Rohan Potta, Nathan Wan, Jed Macapinlac, Bidit Acharyya

4. Deliverables

- **Mobile Responsive Web Application:** A front-end interface built with React and Tailwind CSS that works on both phones and laptops.
- **Backend API and Database:** A server built with Node.js and Express that manages user accounts, room data, and file uploads.
- **AI Knowledge Chatbot:** An integrated AI tool that uses LLM technology to help users search through study notes and find answers to class specific questions.
- **Unit tests for core backend logic, API endpoints, and front-end components** using a JavaScript testing framework (e.g., Vite).
- **Scrum Documentation Suite:** A collection of professional documents including a Project Charter, Product Backlog, Design Document, and three Sprint Reviews.
- **Comparative Experiment Report:** A final analysis of our experience using "vibe coding" versus traditional programming for specific app features.